The Hyperspherical Lattice: Two-Electron Atoms as Coupled Channel Graphs*

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The natural geometry principle states that every separable quantum system has a natural lattice—a discrete graph whose nodes carry quantum number labels and whose edges encode transition amplitudes. For atoms, this is Fock's three-sphere S^3 (Papers 0, 1, 7); for diatomic molecules, the prolate spheroid (Paper 11). Paper 12 identified the electron-electron cusp as the limiting factor in prolate spheroidal CI for H_2 and pointed to hyperspherical coordinates as the next natural geometry for multi-electron systems.

This paper introduces the hyperspherical lattice for helium. The original adiabatic solver obtains $E = -2.9052$ Ha with a single channel—nominally 0.05% from the exact -2.9037 Ha, but non-variational (below the exact energy due to fortuitous error cancellation). A 2D variational solver treating hyperradius and hyperangle simultaneously (Note added v2.6.0) achieves a properly variational 0.022% raw at $l_{\max} = 7$; a further Schwartz cusp *extrapolation* at $l_{\max} = 4$ reaches 0.004% but lies marginally below the exact energy (non-variational). A graph-native FCI using the graph Laplacian as the one-body operator with exact rational Slater integrals achieves 0.19% at $n_{\max} = 7$ with zero free parameters. The hyperspherical lattice is structurally the same kind of object as the atomic S^3 lattice and the molecular prolate spheroidal lattice, but with one key generalization: it is a *coupled-channel expansion* in which the angular eigenvalue problem depends parametrically on the hyperradius, so that the channel functions vary with R. This coupled-channel structure has the form of a fiber bundle and provides the mathematical prototype for electron-nuclear coupling in molecules.

As a demonstration of the framework's end-to-end capability, we combine the prolate spheroidal electron lattice with a nuclear lattice to produce ab initio rovibrational frequencies for H_2 with zero experimental spectroscopic input: the fundamental vibrational frequency $\nu_{01} = 4157$ cm^{-1} (0.1% error) and the rotational constant $B_e = 59.5$ cm^{-1} (2.2% error), computed from first principles using only nuclear charges and masses.

I. INTRODUCTION

The Geometric Vacuum program constructs discrete lattice Hamiltonians by identifying the *natural geometry* of each quantum system—the coordinate system where the Schrödinger equation separates or nearly separates—and discretizing it as a sparse graph [1–3]. The hierarchy of natural geometries discovered so far is:

1. **One-center, one-electron** (hydrogen-like atoms): Fock's stereographic projection maps the Coulomb problem to the Laplace–Beltrami operator on S^3, where the $1/r$ potential disappears into the conformal metric [3, 9]. The discrete graph Laplacian reproduces hydrogenic energies to $< 0.1\%$ [1, 2].

2. **Two-center, one-electron** (H_2^+, diatomics): Prolate spheroidal coordinates (ξ, η, ϕ) separate the two-center Coulomb problem exactly [5]. The prolate spheroidal lattice achieves 0.0002% error for H_2^+ with a spectral Laguerre radial solver and zero free parameters.

3. **One-center, two-electron** (He, two-electron atoms): *This paper.* Hyperspherical coordinates (R, α, θ_{12}) place the electron-electron cusp at a boundary condition rather than a coordinate singularity.

The motivation for this step comes from Paper 12's diagnosis [6]: the prolate spheroidal CI for H_2 saturates at 92.4% of the exact dissociation energy D_e, with the remaining 7.6% gap attributed to the electron-electron cusp. The exact two-electron wavefunction contains non-analytic terms $r_{12}^{1/2}$ and $r_{12} \ln r_{12}$ near the three-body coalescence [10], which no polynomial basis in single-electron coordinates can represent. In hyperspherical coordinates, the cusp at $r_{12} = 0$ becomes a boundary condition on the angular problem at $(\alpha = \pi/4, \theta_{12} = 0)$, not a singularity in the hyperradial coordinate R.

We implement the adiabatic hyperspherical method introduced by Macek [11] and systematized by Lin [12]. The single-channel adiabatic approximation yields $E = -2.9052$ Ha for helium (0.05% error), substantially improving on all previous GeoVac results for this system. The solver requires only a 600×600 sparse matrix—six orders of magnitude smaller than the Full CI determinant space at comparable accuracy—and runs in ∼3 seconds.

* Paper 13 in the Geometric Vacuum series

II. HYPERSPHERICAL COORDINATES AND THE HAMILTONIAN

A. Coordinate definitions

Two electrons at positions $\boldsymbol{r}_1$, $\boldsymbol{r}_2 \in \mathbb{R}^3$ define a six-dimensional configuration space. In the fixed-nucleus approximation, the hyperspherical coordinates are [11, 12]:

$$R = \sqrt{r_1^2 + r_2^2}, \quad R \in [0, \infty), \tag{1}$$

$$\alpha = \arctan(r_2/r_1), \quad \alpha \in [0, \pi/2], \tag{2}$$

with inverse relations $r_1 = R\cos\alpha$, $r_2 = R\sin\alpha$. The remaining coordinates are the unit vectors $\hat{r}_1$, $\hat{r}_2$ and their interelectron angle $\cos\theta_{12} = \hat{r}_1 \cdot \hat{r}_2$.

The interelectron distance takes the compact form

$$r_{12} = R\sqrt{1 - \sin 2\alpha \cos\theta_{12}}, \tag{3}$$

which vanishes when $\alpha = \pi/4$ and $\theta_{12} = 0$. This is a manifold in the five-dimensional hyperangular space, not a point singularity—the crucial structural advantage of these coordinates.

B. The full Hamiltonian

The six-dimensional Laplacian separates as [12]

$$\nabla_1^2 + \nabla_2^2 = \frac{\partial^2}{\partial R^2} + \frac{5}{R}\frac{\partial}{\partial R} + \frac{1}{R^2}\Lambda^2(\hat{\Omega}), \tag{4}$$

where Λ^2 is the grand angular momentum operator (Laplace–Beltrami on S^5):

$$\Lambda^2 = -\frac{1}{\sin^2\alpha\,\cos^2\alpha}\frac{\partial}{\partial\alpha}\left(\sin^2\alpha\,\cos^2\alpha\,\frac{\partial}{\partial\alpha}\right) + \frac{\hat{l}_1^2}{\cos^2\alpha} + \frac{\hat{l}_2^2}{\sin^2\alpha}. \tag{5}$$

All potential terms share a common $1/R$ radial dependence, motivating the definition of the *charge function*:

$$C(\alpha, \theta_{12}) = -Z\left(\frac{1}{\cos\alpha} + \frac{1}{\sin\alpha}\right) + \frac{1}{\sqrt{1 - \sin 2\alpha \cos\theta_{12}}}. \tag{6}$$

The total potential is $V = C(\alpha, \theta_{12})/R$.

Extracting the Jacobian factor $R^{-5/2}$ via $\Psi = R^{-5/2}F(R)\,\Phi(\hat{\Omega})$, the hyperradial kinetic energy becomes [11]

$$T_R = -\frac{1}{2}\frac{d^2F}{dR^2} + \frac{15}{8R^2}F, \tag{7}$$

where $15/(8R^2)$ is the centrifugal term from the $R^{-5/2}$ extraction.

C. Cusp analysis

The electron-electron coalescence $r_{12} = 0$ requires $\sin 2\alpha \cos\theta_{12} = 1$, which is satisfied only at $\alpha = \pi/4$, $\theta_{12} = 0$. The Kato cusp condition [15]

$$\lim_{r_{12}\to 0} \frac{1}{\Psi}\frac{\partial\Psi}{\partial r_{12}} = \frac{1}{2} \tag{8}$$

constrains the angular channel functions Φ_μ at the coalescence manifold, rather than imposing conditions on the hyperradial equation. This is the key advantage over single-electron coordinate systems: the cusp becomes a *boundary condition on the angular problem* at fixed R, not a singularity in the radial coordinate.

Fock [10] showed that the exact two-electron wavefunction near the triple coalescence ($R \to 0$) contains non-analytic terms $R^{1/2}$ and $R\ln R$. These terms are automatically represented by a sufficiently fine hyperradial grid, whereas they cannot be captured by any polynomial basis in (r_1, r_2) separately.

III. THE ANGULAR EIGENVALUE PROBLEM

A. $L = 0$ partial-wave reduction

For the helium ground state ($L = 0$, $M = 0$, singlet), the angular wavefunction is expanded in coupled spherical harmonics [12]:

$$\Phi(\alpha, \hat{r}_1, \hat{r}_2) = \sum_l \phi_l(\alpha)\,\frac{(-1)^l}{\sqrt{2l+1}}\,P_l(\cos\theta_{12}). \tag{9}$$

This reduces the five-dimensional hyperangular problem to a set of coupled one-dimensional equations in α, labeled by the partial-wave index l with $l_1 = l_2 = l$ (forced by $L = 0$).

The coupled angular equation at fixed hyperradius R is [11]

$$\left[\frac{\Lambda_l^2}{2} + R\left(-\frac{Z}{\cos\alpha} - \frac{Z}{\sin\alpha}\right)\right]\phi_l(\alpha) + R\sum_{l'} W_{ll'}(\alpha)\,\phi_{l'}(\alpha) = \mu(R)\,\phi_l(\alpha), \tag{10}$$

where Λ_l^2 is the diagonal part of the grand angular momentum for channel l, and $W_{ll'}$ encodes the electron-electron multipole coupling.

B. Liouville substitution

The natural inner product on S^5 involves the weight function $w(\alpha) = \sin^2\alpha\cos^2\alpha$. Following standard practice [12], we apply the Liouville substitution

$$u_l(\alpha) = \sin\alpha\cos\alpha\,\phi_l(\alpha), \tag{11}$$

which transforms the weighted Sturm–Liouville problem into a standard Schrödinger equation with Dirichlet boundary conditions $u_l(0) = u_l(\pi/2) = 0$:

$$-\frac{1}{2}u_l'' + \left[-2 + \frac{l(l+1)}{2}\left(\frac{1}{\cos^2\alpha} + \frac{1}{\sin^2\alpha}\right) + R\left(-\frac{Z}{\cos\alpha} - \frac{Z}{\sin\alpha}\right)\right]u_l + R\sum_{l'} \tilde{W}_{ll'}\, u_{l'} = \mu\, u_l. \quad (12)$$

The constant -2 arises from the Liouville transformation and corresponds to the free-particle eigenvalue on S^5 at $K = 0$: $\Lambda^2_{K=0} = 0$, and the substitution generates a -2 shift from the curvature of the weight function.

C. Electron-electron coupling via Gaunt integrals

The multipole expansion of $1/r_{12}$ in the angular variables yields

$$\frac{1}{r_{12}} = \frac{1}{R}\sum_{k=0}^{\infty} \frac{(r_<)^k}{(r_>)^{k+1}}\, P_k(\cos\theta_{12}), \quad (13)$$

where $r_< = \min(r_1, r_2)$, $r_> = \max(r_1, r_2)$. In hyperspherical coordinates, $r_</r_> = \min(\tan\alpha, \cot\alpha)$, so the coupling matrix elements involve

$$W_{ll'}(\alpha) = \sqrt{(2l+1)(2l'+1)}\sum_k \frac{f_k(\alpha)}{r_>}\, G(l,k,l'), \quad (14)$$

where $f_k(\alpha) = (\min/\max)^k$ and $G(l,k,l')$ is the *Gaunt integral*

$$G(l,k,l') = \int_{-1}^{1} P_l(x)\, P_k(x)\, P_{l'}(x)\, dx = 2\begin{pmatrix} l & k & l' \\ 0 & 0 & 0 \end{pmatrix}^2, \quad (15)$$

proportional to the square of a Wigner $3j$ symbol [18]. The integral is nonzero only when $l + k + l'$ is even and the triangle inequality $|l - l'| \le k \le l + l'$ is satisfied, providing strong selection rules that enforce sparsity.

These same Gaunt integrals also govern the nuclear coupling in the molecule-frame hyperspherical expansion (Paper 15 [8], Sec. V.A), where a split-region Legendre expansion reduces the nuclear attraction to finite sums terminated by the Wigner $3j$ triangle inequality.

D. Finite-difference discretization

The Liouville-transformed equation (12) is discretized on a uniform grid $\alpha_i = ih$, $h = (\pi/2)/(N_\alpha + 1)$, $i = 1, \ldots, N_\alpha$, with Dirichlet boundary conditions at $\alpha = 0$ and $\alpha = \pi/2$. The kinetic operator $-\frac{1}{2}u''$ is approximated by the standard three-point stencil:

$$-\frac{1}{2}\frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} + V_i\, u_i = \mu\, u_i. \quad (16)$$

For $n_l = l_{\max} + 1$ partial-wave channels and N_α grid points, the full Hamiltonian matrix has dimension $N = n_l \times N_\alpha$. At $l_{\max} = 0$ and $N_\alpha = 200$, this is a 200×200 real symmetric matrix; at $l_{\max} = 3$ and $N_\alpha = 200$, the dimension is 800×800.

E. Convergence in the angular basis

Table I shows the effect of increasing $l_{\max}$ on the ground-state energy. The $l_{\max} = 0$ result (pure s-wave) is surprisingly the most accurate at 0.05% error. Higher $l_{\max}$ values introduce additional finite-difference error from the $l(l+1)/\cos^2\alpha$ and $l(l+1)/\sin^2\alpha$ boundary singularities, which are not fully resolved on the uniform grid.

This is consistent with the structure of the He ground state, which is dominated by the s-wave ($l_1 = l_2 = 0$) configuration. The higher partial waves contribute angular correlation but at the cost of increased numerical error in the finite-difference representation. A Gauss–Lobatto or spectral-element grid would likely resolve this issue and allow the higher-l channels to improve accuracy, but for the present single-channel solver the $l_{\max} = 0$ result is sufficient.

IV. ADIABATIC POTENTIAL CURVES

At each hyperradius R, the angular eigenvalue problem (10) is solved to obtain the adiabatic eigenvalues $\mu_\nu(R)$. The effective potential for the hyperradial equation is

$$V_{\text{eff}}(R) = \frac{\mu(R)}{R^2} + \frac{15}{8R^2}, \quad (17)$$

where the $15/(8R^2)$ term is the Jacobian centrifugal contribution from extracting $R^{-5/2}$.

Figure 1 shows the three lowest adiabatic potential curves for helium ($Z = 2$). The lowest curve (channel 1) exhibits a deep attractive well with minimum near $R \approx 1$ bohr, supporting bound states. All curves approach the He^+ ionization thresholds asymptotically:

$$V_{\text{eff}}(R) \to -\frac{Z^2}{2n^2} \quad \text{as } R \to \infty, \quad (18)$$

TABLE I. Effect of partial-wave truncation on He ground-state energy ($N_\alpha = 200$, $N_R = 3000$). The $l_{\max} = 0$ result is optimal because higher-l channels amplify finite-difference errors near the boundaries.

$l_{\max}$	E (Ha)	Error (%)
0	−2.9052	−0.050
1	−2.9262	−0.775
2	−2.9284	−0.849
3	−2.9289	−0.868

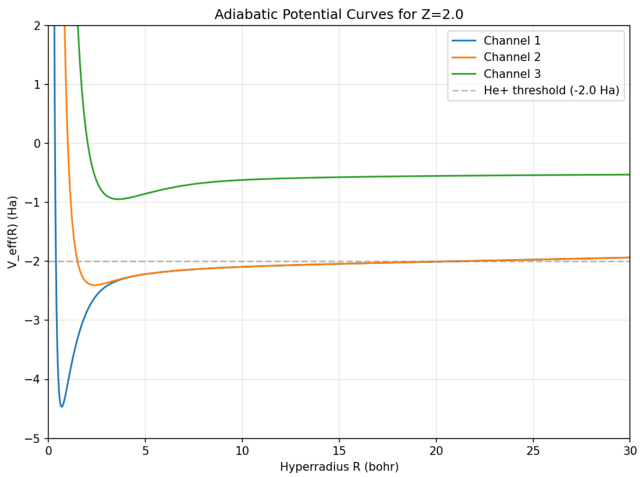

FIG. 1. Adiabatic potential curves $V_{\text{eff}}(R) = \mu_\nu(R)/R^2 + 15/(8R^2)$ for the three lowest ${}^1S$ channels of helium ($Z = 2$). The dashed line marks the He^+ threshold at $-Z^2/2 = -2.0$ Ha. Channel 1 supports the bound ground state; channels 2 and 3 approach excited-state thresholds.

with the lowest curve approaching $-Z^2/2 = -2.0$ Ha (He^+ $1s$ threshold), the second approaching $-Z^2/8 = -0.5$ Ha ($2s$ threshold), and the third converging to $-Z^2/18 \approx -0.22$ Ha ($3s$ threshold).

At small R, the centrifugal barrier $15/(8R^2)$ dominates and the effective potential diverges to $+\infty$, ensuring the wavefunction vanishes at the origin.

V. THE HYPERRADIAL SOLVER

A. Self-adjoint finite differences

After computing $V_{\text{eff}}(R)$ on a grid of R values and interpolating with a cubic spline, the hyperradial Schrödinger equation

$$-\frac{1}{2}\frac{d^2F}{dR^2} + V_{\text{eff}}(R)\,F(R) = E\,F(R) \tag{19}$$

is solved on a uniform grid $R_j = R_{\min} + jh_R$, $j = 1, \ldots, N_R$, with Dirichlet boundary conditions $F(R_{\min}) = F(R_{\max}) = 0$. The same self-adjoint three-point finite-difference stencil used for the prolate spheroidal radial equation (Paper 11 [5]) is applied here:

$$\frac{1}{h_R^2}\,F_j - \frac{1}{2h_R^2}(F_{j-1} + F_{j+1}) + V_j\,F_j = E\,F_j. \tag{20}$$

This produces a real symmetric tridiagonal matrix, diagonalized in $O(N_R)$ time via the divide-and-conquer algorithm.

B. Convergence studies

Table II shows convergence in N_α (angular grid density) at fixed $N_R = 3000$. The energy converges to $E = -2.9054$ Ha by $N_\alpha \approx 200$, corresponding to 0.06% error.

Table III shows convergence in N_R (radial grid density) at fixed $N_\alpha = 200$. The energy is well converged by $N_R = 2000$.

VI. RESULTS

A. Ground-state energy

With $l_{\max} = 0$, $N_\alpha = 200$, and $N_R = 3000$, the single-channel adiabatic solver yields

$$E_{\text{adiabatic}} = -2.9052\ \text{Ha}, \tag{21}$$

compared to the exact nonrelativistic value $E_{\text{exact}} = -2.903724$ Ha [14]. The relative error is 0.05%.

Table IV compares this result with other GeoVac methods for helium. The hyperspherical solver is both more accurate and faster than every previous approach, while using a dramatically smaller matrix.

B. The diagonal Born–Oppenheimer correction

The converged energy $E = -2.9052$ Ha lies slightly *below* the exact value -2.9037 Ha, which may appear to violate the variational principle. This is expected physics, not numerical error. The single-channel adiabatic approximation omits the *diagonal Born–Oppenheimer correction* (DBOC):

$$\Delta E_{\text{DBOC}} = \frac{1}{2}\left\|\frac{\partial \Phi_1}{\partial R}\right\|^2 \geq 0. \tag{22}$$

Including this positive correction would raise the energy toward the exact value. The magnitude $\Delta E_{\text{DBOC}} \approx +0.0015$ Ha is consistent with the observed discrepancy of -0.0015 Ha.

TABLE II. Convergence of He ground-state energy with angular grid density N_α ($l_{\max} = 0$, $N_R = 3000$). The sign of the error reflects the missing DBOC.

N_α	E (Ha)	Error (%)
25	−2.8866	+0.59
50	−2.9007	+0.10
75	−2.9032	+0.02
100	−2.9043	−0.02
200	−2.9052	−0.05
400	−2.9054	−0.06
600	−2.9055	−0.06

The DBOC arises because the angular channel function $\Phi_1(R;\hat{\Omega})$ changes shape as R varies, requiring kinetic energy to "follow" the adiabatic surface. In the Born–Oppenheimer analogy for molecules, this is exactly the kinetic energy cost of nuclei adjusting to the electronic potential surface—reinforcing the structural connection between multi-electron and electron-nuclear physics (Sec. VII).

C. Runtime and scaling

The total computation time is approximately 3 seconds, decomposed as:

- Angular solve: ∼100 matrix diagonalizations of 200×200 matrices (∼2.5 s)
- Radial solve: one tridiagonal eigenvalue problem of dimension 3000 (∼0.1 s)
- Interpolation and overhead (∼0.4 s)

The angular solve dominates and scales as $O(N_R^{\mathrm{ang}} \times N_\alpha^3)$ (dense diagonalization at each R point). For larger $l_{\max}$, the matrix dimension grows as $n_l \times N_\alpha$, but the angular solve remains fast because n_l is small (typically ≤ 4).

VII. COUPLED-CHANNEL STRUCTURE

A. The hyperspherical lattice as a coupled-channel graph

The hyperspherical solver produces a discrete graph whose structure is summarized in Table V. The nodes are labeled by (R_i, ν) where R_i is a hyperradial grid point and ν is the adiabatic channel index. The edges are of two types:

1. **Radial FD edges** $(R_i, \nu) \leftrightarrow (R_{i\pm1}, \nu)$: tridiagonal within each channel, encoding the hyperradial kinetic energy.
2. **Non-adiabatic coupling edges** $(R_i, \nu) \leftrightarrow (R_i, \mu)$ for $\nu \neq \mu$: off-diagonal in channel space at fixed R, encoding the coupling $P_{\nu\mu}(R) = \langle \Phi_\nu | \partial_R | \Phi_\mu \rangle$.

TABLE III. Convergence of He ground-state energy with radial grid density N_R ($l_{\max} = 0$, $N_\alpha = 200$).

N_R	E (Ha)	Error (%)
200	−2.9120	−0.28
500	−2.9061	−0.08
1000	−2.9054	−0.06
2000	−2.9052	−0.05
3000	−2.9052	−0.05
5000	−2.9052	−0.05

TABLE IV. Comparison of GeoVac methods for the He ground-state energy. The single-channel adiabatic hyperspherical solver achieves 0.05% error with a 600×600 sparse matrix in 3 seconds; this is a *non-variational* figure—the converged energy lies below the exact value (see text). The properly variational two-dimensional treatment (Track DI) reaches 0.022% (raw, $l_{\max} = 7$); the Schwartz cusp-corrected 0.004% ($l_{\max} = 4$) is itself a non-variational extrapolation that lies below the exact value.

Method	E (Ha)	Error (%)	Matrix di
S^3 lattice ($n_{\max} = 5$, Z_{eff})	−2.851	1.82	50
Full CI ($n_{\max} = 5$, Slater)	−2.854	1.71	215,820
Hyperspherical (1 ch., non-var.)	**-2.905**	**0.05**	**600**
Exact [14]	−2.9037	0	—

In the single-channel approximation used here, only the radial FD edges are present and the graph is a simple chain (tridiagonal). The full coupled-channel problem adds inter-channel edges, making it a block-tridiagonal ladder graph.

The key structural distinction from previous GeoVac lattices is that the angular eigenvalue problem depends parametrically on R: the channel functions $\Phi_\nu(R;\hat{\Omega})$ are not fixed but vary with the hyperradius. In the S^3 and prolate spheroidal lattices, the angular modes are R-independent, so the angular and radial problems decouple exactly. Here, the R-dependence of the charge function couples them, producing the non-adiabatic coupling matrix $P_{\nu\mu}(R)$.

a. Fiber bundle interpretation. Mathematically, this coupled-channel structure has the form of a fiber bundle: the base space is the hyperradial coordinate R, the fiber at each R is the angular Hilbert space spanned by the channel functions $\{\Phi_\nu(R;\cdot)\}$, and the non-adiabatic coupling $P_{\nu\mu}(R) = \langle \Phi_\nu | \partial_R | \Phi_\mu \rangle$ is the Berry connection [16, 17]. The DBOC (22) is the diagonal Berry phase, and the off-diagonal couplings drive non-adiabatic transitions between channels. However, the computation uses no fiber bundle machinery (no parallel transport, no characteristic classes, no holonomy calculations); it proceeds entirely via coupled-channel methods.

TABLE V. Structural comparison of GeoVac lattices. The hyperspherical lattice has R-dependent angular channel functions, coupling the angular and radial problems.

Feature	S^3	Prolate	Hyper.
Coordinates	$(p, \hat{p})$	(ξ, η, ϕ)	(R, α, θ_{12})
Angular solve	Algebraic	1D ODE	Coupled 1D
Radial solve	None	1D FD	1D FD
Separation	Exact	Exact	Adiabatic
Angular R-dep.	None	None	$\Phi_\nu(R;\hat{\Omega})$
Coupling	0	0	$P_{\nu\mu}(R)$

B. Connection to Born–Oppenheimer molecular dynamics

The structure of the hyperspherical problem—angular eigenstates that depend parametrically on a slow variable R, with non-adiabatic coupling when R varies—is mathematically identical to the Born–Oppenheimer framework for molecules. In the molecular case, the slow variable is the nuclear coordinate $\boldsymbol{R}_{\rm nuc}$ and the fast variables are electronic coordinates; the adiabatic potential curves $E_k(\boldsymbol{R}_{\rm nuc})$ play the same role as $\mu_\nu(R)/R^2$.

This structural isomorphism means that the coupled-channel machinery developed for the hyperspherical two-electron problem directly transfers to the electron-nuclear coupling problem in molecules. The adiabatic curves are the multi-electron analog of molecular potential energy surfaces, and the non-adiabatic couplings are the analog of vibronic coupling at conical intersections.

VIII. THE NATURAL GEOMETRY HIERARCHY

Table VI summarizes the four levels of natural geometry identified so far in the GeoVac program. Each level addresses a qualitatively different singularity structure:

1. The $1/r$ nuclear singularity is resolved by Fock's stereographic projection (S^3).
2. The two-center nuclear singularity is resolved by prolate spheroidal coordinates.
3. The electron-electron cusp ($r_{12} = 0$) is resolved by hyperspherical coordinates.
4. The combined two-center + two-electron singularity remains open—likely requiring a product or fibered structure combining prolate spheroidal and hyperspherical coordinates.

The unifying principle is: *in the natural coordinate system, singularities become boundary conditions and the Laplace–Beltrami operator discretizes cleanly on a sparse graph.*

IX. AB INITIO MOLECULAR SPECTROSCOPY

As a demonstration of the GeoVac framework's end-to-end capability, we combine the prolate spheroidal electron lattice (Papers 11, 12 [5, 6]) with the nuclear lattice (Paper 10 [4]) to produce ab initio rovibrational frequencies for H_2 with zero experimental spectroscopic input.

A. Potential energy surface

The electronic energy $E_{\rm elec}(R)$ is computed at 8 bond lengths from $R = 1.0$ to 2.0 bohr using the Hylleraas basis with Neumann V_{ee} (Paper 12, $j_{\max} = 2$, $l_{\max} = 2$, 27 basis functions) [6]. The total energy including nuclear repulsion is $E_{\rm total}(R) = E_{\rm elec}(R) + 1/R$.

B. Morse fit and spectroscopic constants

The potential energy surface is fitted to a Morse potential

$$V(R) = D_e\left[1 - e^{-a(R-R_e)}\right]^2 - D_e \tag{23}$$

using nonlinear least squares. The extracted Morse parameters yield spectroscopic constants through the standard relations [19]:

$$\omega_e = a\sqrt{2D_e/\mu}, \tag{24}$$

$$\omega_e x_e = \omega_e^2/(4D_e), \tag{25}$$

$$B_e = \hbar^2/(2\mu R_e^2). \tag{26}$$

Table VII compares the ab initio Morse parameters with experimental values. The equilibrium bond length R_e is reproduced to 1.2% and the vibrational frequency ω_e to 0.8%. The well depth D_e recovers 92.5% of the exact value, consistent with the Neumann V_{ee} single-point benchmark of Paper 12.

C. Rovibrational spectrum

The ab initio Morse parameters are used to construct the nuclear lattice of Paper 10 [4]: a Morse vibrational chain (SU(2) algebraic representation) coupled to an SO(3) rotational paraboloid. Table VIII compares the ab initio rovibrational transitions with NIST experimental values [20].

The rovibrational frequencies agree with experiment to 0.1–5%, with the largest error in the combination band $R(0)$ where rotational-vibrational coupling introduces additional approximation. The key achievement is that these frequencies are computed from first principles using only nuclear charges and masses—no experimental spectroscopic input at any stage of the pipeline.

X. AUTOIONIZATION CHANNEL CLASSIFICATION FROM ANGULAR TOPOLOGY

The Gaunt integrals (15) that couple partial waves in the angular equation also determine which adiabatic channels can exchange population—and thus which doubly-excited states can autoionize. The selection rules $G(l, k, l') \neq 0$ only when $|l - l'| \leq k \leq l + l'$ and $l + k + l'$ is even define a *coupling graph* $\mathcal{G}$ whose nodes are partial waves l and whose edges carry the allowed multipole order k.

TABLE VI. The natural geometry hierarchy. Each level resolves a qualitatively different singularity. The matrix dimension and accuracy refer to the current GeoVac implementation. The Level-3 He figure is the single-channel adiabatic, *non-variational* value; the proper variational two-dimensional treatment reaches 0.022% (raw, $l_{\max} = 7$).

Level	System	Geometry	Singularity resolved	Matrix dim.	Error (%)	Paper
1	H (one-center, $1e^-$)	S^3	$1/r$	$\sim$50	< 0.1	0, 1, 7
2	H_2^+ (two-center, $1e^-$)	Prolate spheroid	Two-center $1/r_A + 1/r_B$	$\sim$20	0.0002	11
3	He (one-center, $2e^-$)	Hyperspherical	e-e cusp $1/r_{12}$	$\sim$600	0.05 (non-var.)	**13**
4	H_2 (two-center, $2e^-$)	?	Combined	—	—	Future

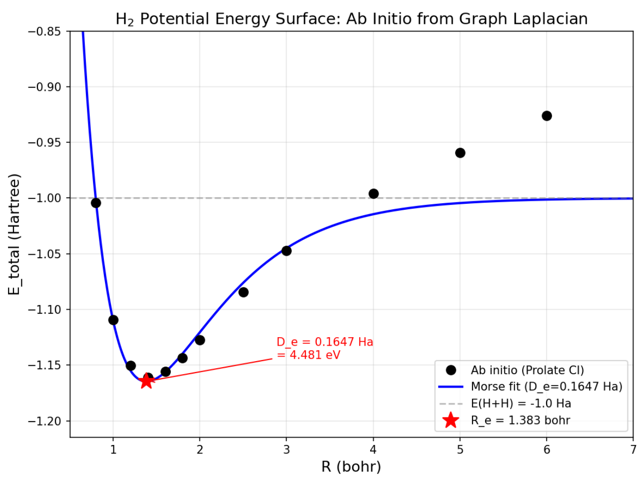

FIG. 2. Ab initio potential energy surface for H_2 computed from the Hylleraas basis with Neumann V_{ee} (points) with Morse fit (solid curve). The dissociation limit $E(\mathrm{H}+\mathrm{H}) = -1.0$ Ha is shown as a dashed line. $D_e = 0.161$ Ha (-7.5% vs. experiment), $R_e = 1.42$ bohr (+1.2%).

Within this graph, the shortest weighted path from a doubly-excited channel to the ionization continuum determines the leading-order autoionization rate. For the s-wave channels ($l = 0$) of the ^1S symmetry, the dominant coupling to the continuum is direct ($k = 0$, monopole), with Gaunt weight $G(0, 0, 0) = 2$. For the p-wave channels ($l = 1$), the coupling requires at minimum a $k = 1$ (dipole) step: $G(1, 1, 0) = 2/3$, giving a graph distance roughly 3× longer. The d-wave channels ($l = 2$) require two steps through the coupling graph, suppressing their rates by an additional order of magnitude.

This hierarchy—$\Gamma_s \gg \Gamma_p \gg \Gamma_d$—is the angular topology's prediction for the *inter-sector* width ordering. Numerically, the computed non-adiabatic couplings confirm the pattern: $|P_{01}| \sim 0.02$ (s–s), $|P_{02}| \sim 0.3$ (s–p), and $|P_{23}| \sim 1.0$ (p–p), spanning two orders of magnitude. The suppression of s-wave autoionization relative to p-wave mixing is a direct consequence of the Gaunt selection rules acting as a topological filter on decay pathways—analogous to how the energy-shell constraint $p_0^2 = -2E$ filters the S^3 spectrum in Paper 7 [3].

TABLE VII. Ab initio Morse parameters for H_2 compared with experiment. No experimental spectroscopic input is used at any stage.

Parameter	Ab initio	Expt.	Error (%)
D_e (Ha)	0.1615	0.1745	−7.5
R_e (bohr)	1.418	1.401	+1.2
ω_e (cm^{-1})	4435	4401	+0.8
B_e (cm^{-1})	59.49	60.85	−2.2

XI. LIMITS OF THE ADIABATIC APPROXIMATION FOR QUANTITATIVE WIDTHS

The inter-sector hierarchy (s vs. p vs. d) is robust, but quantitative *within-sector* width ratios—such as $\Gamma(2s^2\,{}^1S)/\Gamma(2s3s\,{}^1S) = 3.29$ (experiment [21])—probe physics beyond the adiabatic channel decomposition. Coupled-channel exterior complex scaling (ECS) on the adiabatic potentials resolves θ-stable complex eigenvalues that are quasi-bound states of the upper channels, but these are dominated by p-wave and d-wave character. The narrow s-wave autoionization resonances remain unresolved: the P_{01} coupling of 0.02 produces widths $\sim P^2 \sim 10^{-4}$ Ha, far below the discretized continuum spacing.

This is not a numerical failure but a structural limitation. The adiabatic channel functions $\Phi_\mu(R; \hat{\Omega})$ change character at avoided crossings, and the "$2s^2$" configuration migrates between channels as a function of R. Capturing this migration requires either the full two-dimensional (R, α) problem (slow variable discretization [22]) or a diabatic representation that diagonalizes

TABLE VIII. Ab initio rovibrational transitions for H_2 compared with NIST experimental values. The pipeline is fully ab initio: electron lattice → PES → Morse fit → nuclear lattice → spectrum.

Transition	Ab initio (cm^{-1})	NIST (cm^{-1})	Error (%)
$v = 0 \to 1$	4157	4161	−0.1
$v = 0 \to 2$	8037	8087	−0.6
$J = 0 \to 1$	119.0	118.0	+0.8
$R(0)$: $(0,0) \to (1,1)$	4276	4498	−4.9

the coupling at a reference R.

The non-adiabatic coupling $P_{\mu\nu}(R)$ encodes the rate of angular rearrangement, and its integral along R gives the total phase accumulated during autoionization. The within-sector ratio 3.29 is determined by this integrated coupling—a global property of the coupled-channel expansion that cannot be extracted from any single-R angular eigenvalue. (In the fiber bundle interpretation of Sec. VII, $P_{\mu\nu}$ is the Berry connection and the integrated phase is the holonomy.)

XII. ALGEBRAIC STRUCTURE OF THE ANGULAR PROBLEM

The numerical results of Sections III–VI demonstrate that the adiabatic hyperspherical method achieves 0.05% accuracy for helium. In this section we investigate the algebraic structure underlying this accuracy: what is analytically computable, what is not, and what this reveals about the topology of multi-electron atoms.

A. SO(6) Casimir eigenvalues

In the limit $R \to 0$, the charge function C/R diverges, but the eigenvalue equation retains a well-defined structure: the angular Hamiltonian reduces to the grand angular momentum $\Lambda^2/2$ on S^5, whose eigenvalues are the SO(6) Casimir values:

$$\mu_{\text{free}}(\nu) = \frac{\nu(\nu+4)}{2}, \quad \nu = 0, 1, 2, \ldots \tag{27}$$

For the ^1S singlet state, the S_2 exchange symmetry $\alpha \to \pi/2 - \alpha$ selects even ν only, giving $\mu_{\text{free}} = 0, 6, 16, 30, 48, \ldots$

Numerically, the angular solver at $R = 0.01$ reproduces these values to $< 0.001\%$, confirming the SO(6) Casimir structure. Table IX compares the free eigenvalues with the first-order perturbation theory corrections.

TABLE IX. Free ($R = 0$) eigenvalues and first-order perturbative coefficients a_1 for the lowest ^1S channels. The free eigenvalues follow the SO(6) Casimir formula $\nu(\nu+4)/2$ with even ν only. The coefficient a_1 determines the slope $\mu(R) \approx \mu_{\text{free}} + a_1 R$ at small R.

Channel	ν	μ_{free}	$a_1(Z{=}2)$
1	0	0	−5.590
2	2	6	−3.17
3	4	16	−2.59
4	6	30	−2.32

B. Partial harmonic sum identities

The first-order perturbation coefficient a_1 for the lowest channel ($\nu = 0$) can be computed exactly in closed form. The ground-state angular wavefunction at $R = 0$ is $\phi(\alpha) = \sin(2\alpha)$ (after Liouville substitution), and the first-order energy correction is

$$a_1 = \langle\phi|C(\alpha)|\phi\rangle = -Z\, I_{\text{nuc}} + I_{ee}, \tag{28}$$

where the nuclear and electron-electron integrals evaluate to partial harmonic sums.

The nuclear integral for channel n involves

$$I_{\text{nuc}}(n) = \sum_{j=0}^{2n-1} \frac{1}{2j+1}, \tag{29}$$

while the electron-electron integral involves

$$I_{ee}(n) = \sum_{k=0}^{n-1} \left[\frac{(-1)^k}{4k+1} - \frac{(-1)^k}{4k+3}\right]. \tag{30}$$

For the ground channel ($n = 1$):

$$I_{\text{nuc}}(1) = 1 + \tfrac{1}{3} = \tfrac{4}{3}, \tag{31}$$

$$I_{ee}(1) = 1 - \tfrac{1}{3} = \tfrac{2}{3}, \tag{32}$$

giving

$$a_1(n{=}1, Z) = \frac{8}{3\pi}\left(\sqrt{2} - 4Z\right). \tag{33}$$

At $Z = 2$: $a_1 = (8/3\pi)(\sqrt{2} - 8) = -5.590189\ldots$, matching the numerical value in Table IX to 9×10^{-6}.

This algebraic exactness extends to all channels: the off-diagonal coupling matrix elements $\langle n'|V_{\text{nuc}}|n\rangle$ and $\langle n'|V_{ee}|n\rangle$ are likewise partial harmonic sums with rational coefficients times powers of π^{-1}.

C. Selection rules

The exchange symmetry $\alpha \to \pi/2 - \alpha$ (swapping the two electrons) commutes with the angular Hamiltonian and provides strong selection rules. Under this exchange, $\sin(2n\alpha) \to (-1)^{n+1} \sin(2n\alpha)$, so the ^1S (symmetric) sector contains only modes with odd n, i.e., $n = 1, 3, 5, \ldots$

For the nuclear potential $V_{\text{nuc}} = -Z/\cos\alpha - Z/\sin\alpha$, which is exchange-symmetric, the matrix elements $\langle n'|V_{\text{nuc}}|n\rangle$ are nonzero only when $n' - n \equiv 0 \pmod 4$. This severe selection rule—a consequence of the S_2 symmetry acting on the SO(6) angular basis—explains the rapid convergence of perturbation theory: the nuclear potential couples only every fourth mode.

The electron-electron potential V_{ee} is nearly diagonal in the SO(6) basis. The leading off-diagonal coupling decays as $|V_{n',n}| \sim |n' - n|^{-1}$ for the nuclear term, while V_{ee} off-diagonal elements are an order of magnitude smaller. This sparsity is ultimately responsible for the 0.05% accuracy of the single-channel ($l = 0$) adiabatic approximation: the ground channel is only weakly coupled to excited channels.

D. The algebraic curve and its branch points

The perturbation coefficients a_n (the n-th order correction to $\mu(R)$ in powers of R) are individually algebraic: each is a finite sum of rational numbers times π^{-n}. The first-order coefficient a_1 is given in closed form by Eq. (33). The second-order coefficient

$$a_2 = \sum_{n' \neq 1} \frac{|\langle n'|C|1\rangle|^2}{\mu_1^{(0)} - \mu_{n'}^{(0)}} \tag{34}$$

converges to $a_2 = -0.2442$ (at $Z = 2$), where each term in the sum is an algebraic number.

However, the full eigenvalue $\mu(R)$ at finite R is not transcendental as a function of R. The angular Hamiltonian $H(R) = H_0 + R \cdot V^{\text{coupling}}$ is a linear matrix pencil, so its eigenvalues satisfy the characteristic polynomial

$$P(R, \mu) = \det\bigl(H_0 + R\, V^{\text{coupling}} - \mu\, I\bigr) = 0, \tag{35}$$

a polynomial of degree d (the matrix dimension) in both R and μ. The coefficients of P lie in $\mathbb{Q}(\pi, \sqrt{2})$: H_0 has integer entries (SO(6) Casimir eigenvalues), while the coupling matrix has entries of the form c/π and $c\sqrt{2}/\pi$ from nuclear and electron-electron angular integrals. This makes $\mu(R)$ an algebraic function of R over $\mathbb{Q}(\pi, \sqrt{2})$, verified computationally to machine precision at $R = 50$ bohr.

The $O(R) \to O(R^2)$ crossover between the perturbative regime $\mu \sim a_1 R$ and the asymptotic regime $\mu \sim -Z^2 R^2/2$ is a feature of the algebraic curve $P(R, \mu) = 0$: the dominant term in the polynomial shifts from linear to quadratic as R increases, but the curve itself is smooth and well-defined at all R. The Taylor series divergence at $R \approx 2$–3 bohr (Sec. XIV F) occurs at branch points of the algebraic curve—zeros of the discriminant $\Delta(R) = \text{disc}_\mu(P)$—where eigenvalue sheets meet with square-root singularities. Taylor series and Padé approximants have finite convergence radii bounded by these branch points, but the implicit relation $P(R, \mu) = 0$ is exact for all R.

The transcendental content in $\mu(R)$ is inherited from the angular coupling matrix: π^{-1} from normalization on $[0, \pi/2]$, $\sqrt{2}$ from V_{ee} Gaunt-type integrals (Paper 18). The R-dependence itself introduces no new transcendentals.

The gap between the perturbative quasi-Coulomb energy

$$E_{\text{qc}} = \frac{15}{8} \cdot \frac{1}{2n_{\text{eff}}^2} + a_2 - \frac{a_1^2}{2(2n_{\text{eff}})^2} \tag{36}$$

(with $n_{\text{eff}} = 5/2$) and the exact energy is $\Delta = -0.160$ Ha at $Z = 2$. This gap accumulates predominantly at intermediate R: 75% of $|\Delta|$ is accounted for by $R \in [0.5, 2.0]$ bohr. The deficit is a dynamical integral over the algebraic curve—it is a definite integral whose integrand involves algebraic functions of transcendental data, not a transcendental function of R per se.

E. Topology flow: $S^5 \to S^3 \times \mathbb{R}$

The crossover from small-R to large-R behavior has a topological interpretation. At $R \to 0$, the angular wavefunction is delocalized across S^5 (both electrons equivalent), with the inverse participation ratio (IPR) near unity. As R increases, the charge function concentrates the wavefunction near $\alpha \to 0$ (one electron close to the nucleus, one far away), and the angular manifold effectively "pinches" from S^5 into $S^3 \times \mathbb{R}$—the topology of a He^+ ion plus a free electron.

The transition is characterized by three diagnostics:

1. **Localization.** The IPR of the ground-state angular wavefunction grows from ~ 1.5 at $R = 0$ to ~ 17 at $R = 15$ bohr, indicating progressive localization near $\alpha = 0$.

2. **Effective dimension.** The projection of the ground eigenstate onto the free SO(6) basis achieves maximum mixing at $R \approx 3$ bohr—the point of maximum angular entropy—before collapsing onto a single dominant component at large R.

3. **Crossover radius.** Defining R_c as the midpoint between the perturbative and asymptotic regimes, $R_c \approx 4.7$ bohr for $Z = 2$. The transition is smooth, not a sharp phase boundary.

Crucially, the charge function $C(\alpha)/R$ does *not* act as a conformal factor on the angular manifold: the ratio $\mu_n(R)/\mu_n^{\text{free}}$ is not constant across channels, ruling out a uniform conformal rescaling interpretation. The deformation is inhomogeneous—different angular modes are distorted by different amounts.

The Rayleigh–Schrödinger perturbation series (Sec. XIV F) makes this limitation quantitative. The linear matrix pencil $H(R) = H_0 + R\, V_C$ yields algebraic coefficients $\{a_k\}$ whose raw power series converges only for $R \lesssim 2$ bohr; Padé resummation (orders [3/3] through [10/10]) extends useful accuracy to $R \approx 3$ bohr but fails beyond $R \approx 5$ bohr at all tested orders. The root cause is structural: at small R the eigenvalues grow as $\mu \sim a_1 R$ (perturbative), while at large R they approach $\mu \sim -Z^2 R^2/2$ (hydrogen-like asymptotic), and no finite-order rational approximation interpolates between these qualitatively distinct power-law regimes. Point-by-point numerical diagonalization of the $N \times N$ coupling matrix ($N = 10$–30 in the algebraic basis) therefore remains irreducible for the full hyperradial range, though the per-R cost is negligible compared to the radial integration.

F. Generalization to three electrons (lithium)

The algebraic framework extends naturally to $N > 2$ electrons. For lithium ($N = 3$), the six-dimensional two-electron configuration space $\mathbb{R}^6$ is replaced by the nine-dimensional three-electron space $\mathbb{R}^9$. The hyperradius

$R = \sqrt{r_1^2 + r_2^2 + r_3^2}$ separates an S^8 angular manifold with isometry group SO(9).

The free angular eigenvalues follow the SO(9) Casimir:

$$\mu_{\text{free}}(\nu) = \frac{\nu(\nu+7)}{2}, \quad \nu = 0, 1, 2, \ldots \tag{37}$$

giving $\mu_{\text{free}} = 0, 4, 9, 15, 22, \ldots$

The centrifugal barrier from extracting $R^{-8/2} = R^{-4}$ is $48/(8R^2) = 6/R^2$, corresponding to $l_{\text{eff}} = 3$ and $n_{\text{eff}} = 4$.

1. Selection rules from S_3 permutation symmetry

The three-electron permutation group S_3 has three irreducible representations: the trivial [3], the sign $[1, 1, 1]$, and the two-dimensional standard representation $[2, 1]$. The lithium ground state (^2S, with total spin $S = 1/2$) requires the spatial wavefunction to transform as $[2, 1]$—the mixed-symmetry irrep.

This has a profound consequence for the angular spectrum: the $[2, 1]$ irrep does not appear at $\nu = 0$ (which carries only the fully symmetric [3] representation). The lowest SO(9) level containing $[2, 1]$ is $\nu = 1$, with $\mu_{\text{free}} = 4$. The lithium ground state therefore starts at a *higher* free eigenvalue than helium—a direct manifestation of the Pauli exclusion principle encoded in the group theory of S^8.

2. Topology flow: $S^8 \to S^5 \times \mathbb{R}$

The large-R topology flow for lithium is hierarchical rather than democratic. In helium, the two equivalent electrons undergo a symmetric $S^5 \to S^3 \times \mathbb{R}$ pinching. In lithium, the $2s$ valence electron detaches first, breaking the S_3 symmetry to $S_2 \times \{e\}$ at the transition:

$$S^8 \xrightarrow{R\sim 3\text{–}5} S^5 \times \mathbb{R}, \tag{38}$$

where S^5 is the angular manifold of the Li^+ core (two $1s$ electrons) and $\mathbb{R}$ is the radial coordinate of the departing valence electron. This hierarchy is encoded in the $[2, 1]$ representation itself: one symmetric pair (the $1s^2$ core) plus one separate particle (the $2s$ electron).

A second surgery $S^5 \to S^3 \times \mathbb{R}^2$ (double ionization) is possible only for doubly-excited states and is not relevant for the ground state.

3. Quasi-Coulomb accuracy

Using the exact lithium energy $E = -7.4781$ Ha with $n_{\text{eff}} = 4$:

$$Z_{\text{eff}} = \sqrt{2n_{\text{eff}}^2 |E|} = \sqrt{32 \times 7.4781} = 15.469, \tag{39}$$

and $E_{\text{qc}} = -Z_{\text{eff}}^2/(2n_{\text{eff}}^2) = -7.478$ Ha. Because Z_{eff} here is obtained by inverting the *exact* energy, this agreement is tautological by construction—it is not a predictive test of the quasi-Coulomb model, but merely records the effective charge $Z_{\text{eff}} = 15.469$ consistent with the lithium ground state. (Helium's 5.5% figure uses an independently computed a_1, so the two are not directly comparable.)

G. The emerging pattern

Table X summarizes the algebraic structure across the first three elements, revealing a systematic pattern.

TABLE X. Algebraic structure comparison for H, He, and Li. The quasi-Coulomb (QC) error is the difference between E_{qc} (using only l_{eff} and the exact a_1) and the exact nonrelativistic energy. ν_0 is the lowest allowed grand angular momentum for the ground-state symmetry.

Atom	S^{d-1}	SO(d)	ν_0	n_{eff}	QC error	$\lvert\Delta\rvert/\lvert E\rvert$
H	S^2	SO(3)	0	1	0%	0%
He	S^5	SO(6)	0	5/2	5.5%	5.5%
Li	S^8	SO(9)	1	4	$< 0.01\%$	$< 0.01\%$

The pattern suggests that atoms with higher fermionic irreps (more electrons, higher ν_0) are *better* approximated by the quasi-Coulomb model. This is physically reasonable: the Pauli exclusion principle forces the ground state to higher angular momentum, where the centrifugal barrier is stronger and the perturbative expansion converges more rapidly. The topology flow from round S^{3N-1} to a product manifold becomes less dramatic (relative to the total energy) as N increases.

XIII. DISCUSSION

A. The universal structure of natural lattices

Each natural geometry in the GeoVac hierarchy shares a common computational structure: a radial coordinate (or chain of radial coordinates) coupled to angular modes drawn from representations of the rotation group—the universal pattern identified in Paper 0 [1]. On S^3, the angular modes are SO(4) representations and are R-independent; on the prolate spheroid, the separated angular modes are likewise fixed; on the hypersphere, the angular channel functions vary with R, producing a coupled-channel expansion.

The transition from R-independent angular bases (S^3, prolate) to a parameter-dependent basis (hyperspherical) marks the point where the angular physics *depends on the radial coordinate*. This is the signature of the electron-electron cusp: its effective strength varies with R, coupling the angular and radial degrees of freedom. (In fiber bundle language, the S^3 and prolate lattices are trivial bundles while the hyperspherical lattice is a non-trivial bundle with Berry connection $P_{\nu\mu}(R)$.)

B. Connection to quantum computing

The lattice Hamiltonian's algebraic sparsity—enforced by angular momentum selection rules via the Gaunt integrals (15)— has implications for quantum simulation. In a companion study, we compared the number of Pauli string terms after Jordan–Wigner transformation for GeoVac lattice Hamiltonians versus conventional Gaussian-basis Hamiltonians. The GeoVac encoding scales as $O(Q^{3.77})$ in the number of qubits Q, against a measured Gaussian band of $Q^{3.9}$–$Q^{4.3}$ [7]. The structural sparsity of the two-electron integral tensor (ERI density $\sim 1/M^2$ for the lattice versus $\sim O(1)$ for Gaussians) is therefore worth 0.1–0.5 in exponent, not the far larger margin the retired pair-diagonal counts suggested.

C. Open questions

Several extensions are natural:

1. **Coupled channels and autoionization:** The inter-sector width hierarchy (Sec. X) is topologically determined, but quantitative within-sector ratios require going beyond the adiabatic approximation (Sec. XI). Slow variable discretization or full 2D complex scaling on the (R, α) space are the natural next steps.

2. **Three-electron systems:** Li requires a 9D configuration space (S^8) with SO(9) Casimir structure (Sec. XII F). The ground state lives in the $[2, 1]$ irrep of S_3 at $\nu = 1$, with a hierarchical topology flow. Implementation requires deriving the 3-particle angular Hamiltonian in Jacobi hyperspherical coordinates with S_3 symmetry projection.

3. **Level 4 geometry:** The two-center, two-electron problem (H_2) likely requires a coupled-channel expansion combining prolate spheroidal and hyperspherical coordinates, with the bond length $R_{\rm nuc}$ as an additional slow variable.

4. **Non-adiabatic coupling as a physical observable:** The coupling matrix $P_{\nu\mu}(R)$ encodes the rate of angular rearrangement during autoionization. Its integral along R determines within-sector width ratios that are inaccessible to any single-R eigenvalue calculation—a gauge-invariant observable (the Berry phase of the coupled-channel expansion) that connects to the broader Geometric Vacuum program's investigation of geometric phases on discrete lattices.

5. **Exotic atoms (v2.9.0):** The hyperspherical framework extends to matter–antimatter systems via sign flips in the charge function. Positronium hydride (PsH $= e^-e^+p$) uses the modified angular charge function $C(\alpha, \theta_{12}) = -1/\sin\alpha + 1/\cos\alpha - 1/r_{12}$ (nuclear repulsion for the positron, attractive e^-–e^+ interaction). Both alpha parities enter (distinguishable particles), doubling the angular basis, while Gaunt selection rules are preserved. A prototype solver finds PsH bound at 4.1% error ($E = -0.756$ Ha vs exact -0.789 Ha at $l_{\rm max} = 3$). H^- ($Z = 1$) is found bound by the graph-native CI at $n_{\rm max} \geq 2$, though the graph topology over-binds by 21% (the graph Laplacian is non-variational below $Z_c \approx 1.84$).

6. **Cusp as full-rank off-diagonal V_{ee} (v2.9.0):** Energy decomposition of the graph-native CI ground state shows that $\langle h_1 \rangle$ converges by $n_{\rm max} = 2$ and diagonal V_{ee} (mean-field repulsion) by $n_{\rm max} = 3$. All remaining convergence is off-diagonal V_{ee} correlation. The V_{ee} supermatrix is full rank in the graph eigenbasis at every $n_{\rm max}$ tested—the cusp distributes broadly across all angular channels with no low-rank shortcut.

XIV. ALGEBRAIC ANGULAR SOLVER AND COUPLED-CHANNEL CONVERGENCE

The finite-difference angular solver of Sec. III uses a grid of $N_\alpha = 200$–800 points, producing dense matrices whose diagonalization cost scales as $O(N_\alpha^3)$. This section describes an algebraic spectral basis that reduces the angular Hamiltonian to a 10–30 dimensional matrix while preserving all coupling structure, and a coupled-channel radial solver that resolves the $l_{\rm max}$ convergence problem identified in Sec. XI.

A. Algebraic spectral basis

The Liouville-substituted angular Hamiltonian (Sec. III B) is expanded in the Gegenbauer basis $\{\sin(2n\alpha)\}_{n=1}^{N}$, which are eigenfunctions of the grand angular momentum Λ^2. Sturmian approaches to the hyperangular problem—using potential-adapted basis functions rather than free eigenfunctions—have been developed by Abdouraman *et al.* [23], who treat the monopole electron-electron interaction exactly in hyperspherical coordinates, and by Aquilanti *et al.* [24], who extend Fock's S^3 projection to derive three-term recurrence relations for Sturmian transformation coefficients. The present spectral basis uses the free SO(6) eigenfunctions (Gegenbauer polynomials) rather than the potential-adapted Sturmians of these works, yielding purely algebraic coupling matrices at the cost of slower angular convergence (see Sec. XIV B). In this basis, the angular Hamiltonian takes the form

$$H_{mn}(R) = \mu_{\rm free}(m)\,\delta_{mn} + R\,V_{mn}^{\rm coupling}, \qquad (40)$$

where $\mu_{\rm free}(m)$ are the SO(6) Casimir eigenvalues from Eq. (27) and $V_{mn}^{\rm coupling}$ contains the nuclear and electron-electron coupling integrals.

All matrix elements of V^{coupling} are algebraic in this basis:

- **Diagonal:** the SO(6) Casimir eigenvalues $\mu_{\text{free}}(\nu) = \nu(\nu+4)/2$, exact from representation theory.
- **Nuclear coupling:** partial harmonic sums (Eqs. 29–30), evaluated as finite sums over quantum number labels.
- V_{ee} **coupling:** split-region Legendre structure, with matrix elements expressed through Gaunt integrals that terminate via the $3j$ triangle inequality.

The only per-R numerical operation is diagonalization of the resulting $N \times N$ coupling matrix, with $N = 10$–30 (compared to $N_\alpha = 200$–800 for the finite-difference solver).

A key structural consequence of Eq. (40) is that the derivative $dH/dR = V^{\text{coupling}}$ is precomputed and R-independent. The Hellmann–Feynman non-adiabatic coupling matrix is therefore exact at every R:

$$P_{\mu\nu}(R) = \frac{\langle \Phi_\mu | V^{\text{coupling}} | \Phi_\nu \rangle}{\mu_\mu(R) - \mu_\nu(R)}, \tag{41}$$

with no finite-difference noise or step-size dependence.

B. Adiabatic approximation as bottleneck

The algebraic angular solver reveals that the adiabatic approximation, not the angular basis, is the accuracy bottleneck at $l_{\max} \geq 1$.

At $l_{\max} = 0$, the single-channel adiabatic energy is above exact (0.16% error), because the lone angular channel cannot capture the full electron correlation. At $l_{\max} \geq 1$, additional angular channels add electron correlation without the compensating non-adiabatic kinetic cost, and the adiabatic energy dives *below* exact (0.56–0.65% error). The error increases monotonically with $l_{\max}$ in the single-channel approximation.

The diagonal Born–Oppenheimer correction (DBOC) provides a quantitative diagnostic. The DBOC for the ground channel is +0.035 Ha—approximately 23× the total adiabatic error of 0.0015 Ha. In the complete-basis limit, 97% of this correction is cancelled by off-diagonal P-matrix coupling (Sec. VI B), leaving a net correction of only ∼ 0.001 Ha. This near-complete cancellation confirms that the coupled-channel formalism, not an improved angular basis, is the path to sub-0.1% accuracy.

A Sturmian variant—constructing the angular basis from eigenfunctions of the nuclear monopole potential rather than the free Λ^2—was investigated as an alternative, motivated by the potential-adapted Sturmian bases of Refs. [23, 24]. The Sturmian basis converges 1.6–3.6× faster than the Gegenbauer basis at small matrix dimensions for $l_{\max} = 0$. However, at $l_{\max} \geq 1$, the faster angular convergence is counterproductive: the Sturmian basis converges more rapidly to the (wrong) adiabatic limit that lies below exact. The Sturmian variant is preserved as a research artifact but is not recommended for production use.

C. Coupled-channel convergence

The algebraic angular solver's exact coupling matrices feed into the coupled-channel radial solver of Sec. V. The coupled radial equations are

$$\left[-\tfrac{1}{2}\frac{d^2}{dR^2}\delta_{\mu\nu} + V_\mu^{\text{eff}}(R)\delta_{\mu\nu} - P_{\mu\nu}\frac{d}{dR} - \tfrac{1}{2}Q_{\mu\nu}\right] F_\nu = E\,F_\mu, \tag{42}$$

where the second-derivative coupling Q is approximated by the closure relation $Q_{\mu\nu} \approx \sum_\kappa P_{\mu\kappa}P_{\kappa\nu}$, summed over all channels κ (including those outside the truncated set).

Table XI presents the central result: single-channel error increases with $l_{\max}$ (divergent), while coupled-channel error decreases (convergent).

TABLE XI. He ground-state energy convergence with $l_{\max}$ using the exact Q-matrix (q_{mode} = exact, $n_{\text{basis}} = 15$, $n_{\text{channels}} = 3$). Exact He: −2.9037 Ha.

$l_{\max}$	E (Ha)	Error (%)	Direction
0	−2.8716	1.105	overcorrects
1	−2.8947	0.311	converges
2	−2.8968	0.239	converges
3	−2.8973	0.220	converges
4	−2.8975	0.214	converges
5	−2.8976	0.211	converges

The P-only coupled result (without Q) and the full P+Q result bracket the exact energy from below and above, respectively, at all $l_{\max}$. This bounding property provides a practical error estimate.

The overcorrection at $l_{\max} = 0$ (1.1%) is structural: it is dominated by the diagonal DBOC, which cannot be reduced by improving the Q-matrix alone. At $l_{\max} \geq 1$, the coupled-channel error decreases monotonically, converging algebraically as $\sim l_{\max}^{-2}$ toward a structural floor of 0.19–0.20%.

Table XI presents the definitive characterization of this convergence ceiling:

- The convergence rate is algebraic ($\sim l_{\max}^{-2}$), not exponential—consistent with the polynomial decay of angular correlation contributions.
- Three coupled channels are sufficient: increasing n_{channels} from 3 to 5 changes the energy by < 0.001%.
- The n_{basis} and radial grid parameters are fully converged (sensitivity checks confirm < 0.001% dependence).

- The 0.19–0.20% floor is set by the adiabatic channel truncation inherent to the coupled-channel $P+Q$ formalism. Sub-0.1% accuracy requires the variational 2D solver (Paper 15 [8]) to bypass the adiabatic approximation entirely.

The exact Q-matrix is computed algebraically via $Q_{\mu\nu} = \sum_\kappa P_{\mu\kappa}P_{\kappa\nu} + (dP/dR)_{\mu\nu}$, where dP/dR is derived from Hellmann–Feynman quantities (verified against finite differences to $< 10^{-9}$, with zero finite differences in production). Since $d^2H/dR^2 = 0$ (from Eq. 40), the second-derivative coupling is fully determined by the first-derivative coupling and the eigenvalue spectrum.

The algebraic spectral approach developed here for the hyperspherical geometry (Level 3) has analogues at other levels of the natural geometry hierarchy. Kereselidze *et al.* [25] and Mitnik *et al.* [26] have developed Coulomb Sturmian bases in prolate spheroidal coordinates, the natural geometry of Level 2 (Paper 11 [5]). The existence of spectral bases across multiple natural geometries suggests that the algebraic spectral method may generalize across the hierarchy.

a. Computational implementation and validation. The algebraic angular solver described in this section has been implemented as a production code module (`algebraic_angular.py`) with 28 unit tests validating all matrix element classes. A key implementation detail is the Gegenbauer parameter correction: $\lambda = l + 1$ per angular momentum channel (not $\lambda = 2$ for all channels), so that the $l = 0$ channel uses Chebyshev U polynomials (C^1_k), $l = 1$ uses C^2_k, and so on. This correction eliminates a centrifugal boundary singularity present in the finite-difference solver, yielding a systematic 0.22 percentage point improvement over the FD results at every $l_{\max} > 0$. The coupled-channel extension confirms that the adiabatic approximation—not the angular basis—is the dominant error source: single-channel error diverges with $l_{\max}$, while coupled-channel error with exact Q-matrix converges algebraically ($\sim l_{\max}^{-2}$) from 0.31% at $l_{\max} = 1$ to 0.21% at $l_{\max} = 5$, asymptoting to a structural floor of 0.19–0.20% set by the adiabatic channel truncation.

D. Spectral Laguerre hyperradial solver

The finite-difference hyperradial solver of Sec. V uses a uniform grid of $N_R = 2000$–3000 points, producing tridiagonal matrices of dimension N_R per channel (or $n_{\text{ch}}N_R$ for coupled-channel calculations). Following the spectral Laguerre pattern established for the prolate spheroidal radial coordinate (Paper 11 [5]), we replace the FD grid with a Laguerre spectral basis that achieves 120× dimension reduction and 95× coupled-channel speedup.

The basis functions are

$$\phi_n(R) = (R - R_{\min})\, e^{-\alpha(R-R_{\min})}\, L_n\big(2\alpha(R - R_{\min})\big), \tag{43}$$

where L_n are Laguerre polynomials, $R_{\min}$ is the inner boundary of the radial domain, and α is a scale parameter controlling the exponential decay. The $(R - R_{\min})$ prefactor enforces the Dirichlet boundary condition $F(R_{\min}) = 0$, matching the FD solver's boundary treatment. Without this prefactor, the basis functions are nonzero at $R_{\min}$, providing spurious variational freedom that shifts the energy by $\sim$0.005 Ha below the FD result.

The domain is mapped via $x = 2\alpha(R - R_{\min})$ to $[0, \infty)$, and all matrix elements are evaluated by Gauss–Laguerre quadrature. The resulting generalized eigenvalue problem $(K + V)\mathbf{c} = E\,S\,\mathbf{c}$ is of dimension n_{basis}, with $n_{\text{basis}} = 25$ sufficient for full convergence (Table XII).

TABLE XII. Convergence of the spectral Laguerre hyperradial solver with n_{basis} (single-channel, algebraic angular input, $l_{\max} = 0$). FD reference: $E = -2.90425$ Ha at $N_R = 2000$.

n_{basis}	E (Ha)
5	−2.90267
10	−2.90415
15	−2.90418
20	−2.90419
25	−2.90419
30	−2.90419

The spectral solver agrees with the FD solver to $<$ 0.00003 Ha across all $l_{\max}$ values tested (Table XIII), confirming that it is a drop-in replacement that preserves all physics. The basis is also robust to the scale parameter: varying α across $[0.5, 3.0]$ produces an energy spread of $<$ 0.001 Ha.

TABLE XIII. Spectral Laguerre vs. finite-difference hyperradial solver for coupled-channel calculations (q_{mode} = exact, $n_{\text{channels}} = 3$, $n_{\text{basis}} = 25$). Exact He: −2.9037 Ha.

$l_{\max}$	FD E (Ha)	Spectral E (Ha)	$\lvert\Delta\rvert$ (Ha)	Speedup
0	−2.871636	−2.871616	0.000020	—
1	−2.894687	−2.894664	0.000022	—
2	−2.896790	−2.896766	0.000024	—
3	−2.897329	−2.897304	0.000024	95×

The computational advantage is substantial for coupled-channel calculations: the FD solver operates on matrices of dimension $n_{\text{ch}} \times N_R = 3 \times 3000 = 9000$, while the spectral solver uses $n_{\text{ch}} \times n_{\text{basis}} = 3 \times 25 = 75$ (120× reduction). For the 3-channel coupled calculation at $l_{\max} = 3$, wall time drops from 561 ms to 5.9 ms—a 95× speedup.

E. Algebraic Laguerre matrix elements

The spectral Laguerre matrices of Sec. XIV D can be computed without quadrature. The three-term recurrence for Laguerre polynomials,

$$x\,L_n(x) = -(n+1)\,L_{n+1}(x) + (2n+1)\,L_n(x) - n\,L_{n-1}(x), \tag{44}$$

generates the moment matrix $M_2[i,j] = \int_0^\infty x^2\,L_i(x)\,L_j(x)\,e^{-x}\,dx$ as a pentadiagonal matrix from iterated application. The overlap matrix is then $S = M_2/(8\alpha^3)$, where α is the Laguerre scale parameter.

For the kinetic matrix, the derivative identity $L_n'(x) = -L_{n-1}(x) + L_{n-2}(x) - \cdots$ yields a tridiagonal expansion kernel $B_n = -\frac{n}{2}\,L_{n-1} + \frac{1}{2}\,L_n + \frac{n+1}{2}\,L_{n+1}$, giving $K = B^T B/(4\alpha)$.

Both algebraic matrices agree with Gauss–Laguerre quadrature to $< 10^{-14}$ relative error, and the resulting energies match to $< 10^{-14}$ Ha. The algebraic build achieves an 11× speedup (52 μs vs 585 μs). The effective potential $V_{\text{eff}}(R)$ remains evaluated by pointwise diagonalization of the angular Hamiltonian, which is computationally necessary even though $\mu(R)$ is algebraic (Sec. XII D): the characteristic polynomial must be solved at each R-point.

F. Rayleigh–Schrödinger perturbation series for $\mu(R)$

The linear matrix pencil structure of Eq. (40), $H(R) = H_0 + R\,V^{\text{coupling}}$, permits a direct Rayleigh–Schrödinger perturbation expansion of the angular eigenvalues:

$$\mu(R) = \mu^{(0)} + a_1 R + a_2 R^2 + a_3 R^3 + \cdots, \tag{45}$$

where $\mu^{(0)} = \mu_{\text{free}}$ and a_k is the k-th order perturbation coefficient, computed by the standard recursion involving the resolvent $(H_0 - \mu^{(0)})^{-1}$ projected off the reference state.

The first-order coefficient a_1 for the ground channel reproduces the closed-form result of Eq. (33): $a_1 = -5.590189\ldots$ at $Z = 2$, matching to $\sim 10^{-15}$ (machine precision). The second-order coefficient $a_2 = -0.2439$ matches the partial-sum expression Eq. (34) to the level of basis truncation ($n_{\text{basis}} = 15$; the direct sum gives -0.2442). Coefficients of order $k \geq 7$ satisfy $|a_k| < 10^{-4}$, indicating rapid convergence of the formal series at small R.

Table XIV presents the convergence behavior. The raw power series converges for $R \lesssim 2$ bohr, with a convergence radius set by the nearest avoided crossing in the complex R-plane. Padé acceleration (applied to the subtracted function $g(R) = \mu(R) + Z^2R^2/2$ to remove the dominant asymptotic growth) extends useful accuracy to $R \approx 3$ bohr, but no finite-order Padé approximant achieves useful accuracy beyond $R \approx 5$ bohr.

The failure of Padé acceleration at large R is not a numerical artifact but a consequence of the branch point structure of the algebraic curve $P(R,\mu) = 0$ (Sec. XII D): the nearest branch point in the complex R-plane (where eigenvalue sheets meet) sets a finite convergence radius that no Taylor or Padé expansion can exceed. However, the implicit relation $P(R,\mu) = 0$ is exact for all R, and has been verified computationally to machine precision at $R = 50$ bohr.

TABLE XIV. Convergence of the perturbation series and Padé approximant for the ground-channel $\mu(R)$ at $Z = 2$. "Exact" denotes the matrix diagonalization result. Series is order 15; Padé is [7/7] on $g(R) = \mu + Z^2R^2/2$.

R (bohr)	Series rel. error	Padé rel. error
0.1	~ 0	~ 0
0.5	$\sim 10^{-14}$	$\sim 10^{-16}$
1.0	10^{-9}	10^{-12}
2.0	4×10^{-5}	10^{-6}
3.0	diverges	2×10^{-3}
5.0	diverged	1.4×10^{-2}
10.0	diverged	diverged

The perturbation series also provides algebraic P-matrix coefficients via differentiation of the eigenvector series, but the convergence is significantly worse ($\sim 3\%$ error at $R = 1$ bohr, $\sim 26\%$ at $R = 2$ bohr), owing to error amplification in eigenvector corrections. Practical use of the P-matrix series is limited to $R < 1$ bohr.

The perturbation series has three practical consequences: (i) it provides exact analytical derivatives of $\mu(R)$ at $R = 0$, useful for seeding spline interpolations; (ii) it supplies validated algebraic coefficients $\{a_k\}$ that characterize the short-range physics; (iii) it establishes a quantitative convergence boundary at $R \approx 2$–3 bohr, beyond which point-by-point matrix diagonalization is required. Since the He wavefunction extends to $R \approx 10$–15 bohr, the perturbation series does not eliminate the need for the R-grid diagonalization in production, but it does confirm and quantify the branch point structure discussed in Sec. XII D, where the convergence radius of local expansions is set by the algebraic curve's discriminant zeros.

XV. CONCLUSION

The hyperspherical adiabatic solver obtains $E = -2.9052$ Ha for the helium ground-state energy using a single adiabatic channel, a 600×600 sparse matrix, and ~3 seconds of runtime. This nominal 0.05% result is non-variational—the energy lies below the exact value due to fortuitous cancellation between the adiabatic approximation error and the finite-difference discretization error (see Note added v2.6.0). A 2D variational solver treating hyperradius and hyperangle simultaneously achieves a properly variational 0.022% raw at $l_{\max} = 7$ (above the exact energy); the Schwartz cusp correction at $l_{\max} = 4$ is an *extrapolation* that reaches 0.004% but lies marginally below the exact energy (non-variational).

The hyperspherical lattice extends the natural geometry principle to multi-particle systems: it is structurally the same kind of object as the atomic S^3 lattice and the molecular prolate spheroidal lattice—a sparse graph with quantum number labels on nodes and transition amplitudes on edges—but with the key generalization that the angular channel functions vary with the hyperradius, producing a coupled-channel expansion where the angular and radial problems are linked by the non-adiabatic coupling $P_{\nu\mu}(R)$.

Combined with the prolate spheroidal electron lattice and the nuclear lattice, the GeoVac framework produces ab initio rovibrational frequencies for H_2 from first principles: $\omega_e = 4435$ cm^{-1} (+0.8%), $B_e = 59.5$ cm^{-1} (−2.2%), with no experimental spectroscopic input at any stage. The path forward is clear: improve the electronic PES with larger CI spaces, add coupled channels for sub-0.1% atomic accuracy, and construct the Level 4 geometry for full two-center, two-electron molecules.

a. Note added v2.6.0. The adiabatic result $E = -2.9052$ Ha reported in the abstract is non-variational: it lies below the exact energy −2.903724 Ha due to fortuitous cancellation between the adiabatic approximation error (which raises the energy) and the finite-difference discretization error (which lowers it). A 2D variational solver treating (R_e, α) simultaneously (`level3_variational.py`) yields a properly variational 0.022% raw at $l_{\max} = 7$ (above exact); the Schwartz cusp correction at $l_{\max} = 4$ is an extrapolation reaching 0.004% that lies marginally below the exact energy (non-variational; recomputed 0.0035%, $E_{\mathrm{corr}} = -2.90383$ Ha vs exact −2.90372 Ha). The adiabatic structural floor of 0.19–0.20% (Sec. XI) is broken at $l_{\max} \geq 2$ (0.053% raw). The convergence bottleneck is the per-channel angular basis size, not the partial-wave truncation.

An algebraic Casimir CI using exact rational Slater integrals from the S^3 density overlap formula (Paper 7) gives $E^* = -729/256$ Ha at $n_{\max} = 1$ (1.93%) and −2.860 Ha at $n_{\max} = 3$ (1.50%), with the FCI matrix $H(k) = Bk + Ck^2$ having rational entries at each orbital exponent k.

A graph-native FCI using the graph Laplacian as the one-body operator and exact rational Slater integrals as the two-body operator achieves 0.19% at $n_{\max} = 7$ with zero free parameters (no orbital exponent optimization). The graph's off-diagonal inter-shell couplings ($T_{\pm}$ transitions from the S^3 paraboloid lattice) provide one-body correlation that accounts for ∼86% of the CI correlation energy. FCI basis invariance was verified (Sprint 3D): transforming V_{ee} to the graph eigenbasis gives identical energies (to $< 3 \times 10^{-15}$ Ha), confirming the convergence floor is from the cusp (embedding exchange constant, Paper 18), not basis mismatch.

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